

“Contract” Table

GET /api/products

1. Overview

Field	Detail
Endpoint	GET /api/products
Who sends it	Browser (Client)
Who handles it	Express.js Server
Data source	JSON file (./data/products.json)
Purpose	Fetch product data dynamically and display on the webpage
Response format	JSON

2. Input — Query Parameter

Parameter	Type	Required	Default	Description	Example
category	string	Optional	all	Filter products by category; return all if omitted	?category=Running shoes for men

3. JSON Data Structure (Source File)

File path: ./data/products.json

Field	Type	Description	Example
id	number	Unique product ID	1
image	string	Relative path to product image	"images/card-item1.jpg"
title	string	Product name	"AeroSwift Pro Runner"
price	number	Product price (USD)	120
category	string	Product category label	"Running shoes for men"

Available categories: - Running shoes for men - Casual shoes for men
- Walking shoes - Outdoor footwear - Training shoes - Formal shoes -
Athletic shoes - Casual footwear - Boots

4. Output — Success Response 200 OK

Field	Type	Description
success	boolean	true when request succeeds
data[]	array	List of product objects (filtered or all)
data[].id	number	Unique product ID
data[].image	string	Relative path to product image
data[].title	string	Product name
data[].price	number	Product price
data[].category	string	Product category

Example (with ?category=Boots):

```
{
  "success": true,
  "data": [
    {
      "id": 9,
      "image": "images/card-item9.jpg",
      "title": "Rugged Terrain Boot",
      "price": 160,
      "category": "Boots"
    }
  ]
}
```

5. Output — Error Response

Status	When it happens
404	JSON file not found or cannot be read
500	Server error while reading or filtering the file

Example:

```
{
  "success": false,
```

```

    "error": "FILE_NOT_FOUND",
    "message": "Failed to load products data"
  }
}

```

6. Actors

Actor	Role
Browser	Calls <code>requestProducts()</code> → <code>fetch()</code> → receives JSON → calls <code>renderProducts()</code>
Express Server	Receives request, reads JSON file, filters by category, returns response
JSON File	Flat file data source (<code>./data/products.json</code>) — no database needed

7. Data Flow

Page Load → API Request → Read JSON File → Filter → UI Render

Step	From	To	Message / Action
	Browser	Express	<code>requestProducts()</code>
	Browser	Express	<code>fetch('/api/products?category=?')</code>
	Express	JSON	<code>fs.readFile('./data/products.json')</code>
	JSON	Express	Return full products array
	Express	Express	Filter by <code>category</code> if param is provided
	Express	Browser	Return <code>{ success: true, data[] }</code>
	Browser	—	<code>renderProducts(data)</code> → update <code>#product-container</code>
